HAT-P-54b

R-1568

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-54_190131` · created 2026-08-01T22:16:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458514.62 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.057
Transit depth (Rp/Rs)^2	3.63 +/- 2.16 [%]
Orbital Inclination (inc)	89.74 +/- 2.27
Airmass coefficient 1 (a1)	0.25 +/- 0.12
Airmass coefficient 2 (a2)	1.17 +/- 1.0
Scatter in the residuals of the lightcurve fit is	43.03 %
Best Comparison Star	#2 - [614, 302]
Optimal Aperture	3.31
Optimal Annulus	5.39
Transit Duration (day)	0.0541 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.057 vs 0.1572 (deviation 0.49σ)
Duration fit vs published	0.0541 d vs 0.075 d (deviation 4.98σ)
Mid-time O–C	1.1 min
Depth SNR	2.8
Residual scatter	43.03 %

Light curve

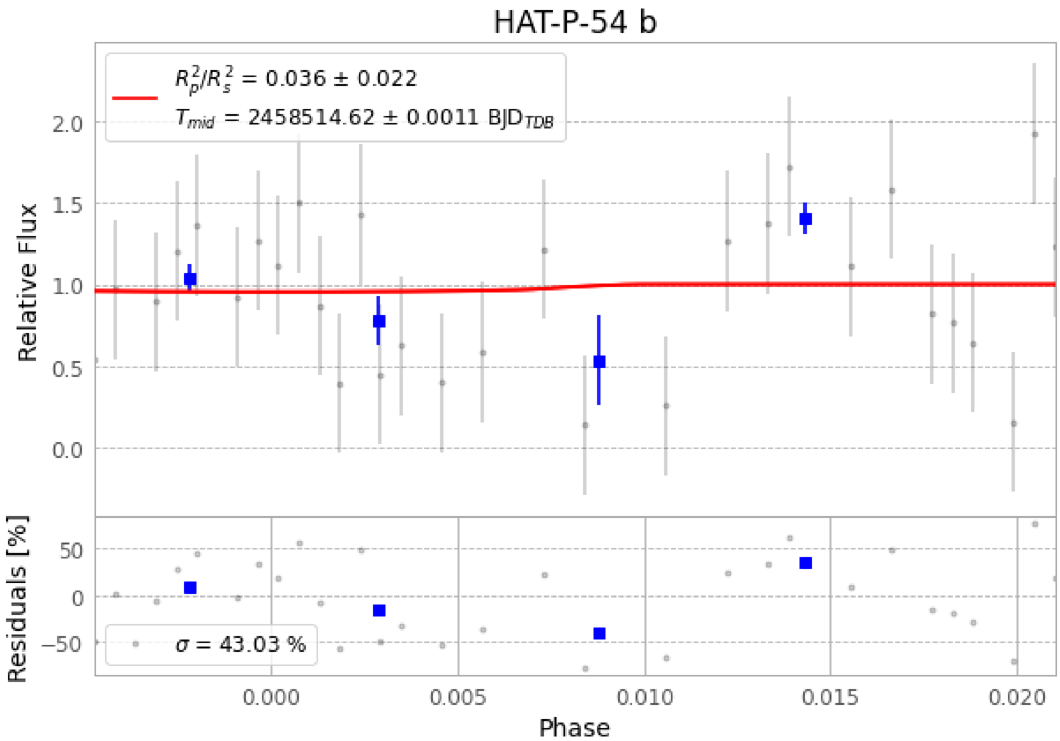

Figure 1 — FinalLightCurve_HAT-P-54 b_2019-01-30.png (SHA-256 aca2a53d952f3a1b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [586, 238], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e4733c42e7c97ebd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

48 FITS frames (32 MB), `HATP-541901310218l2.FITS` → `HATP-541901310439l2.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-190131.log (01085fa5e439…), FinalLightCurve_HAT-P-54 b_2019-01-30.png (aca2a53d952f…), FinalLightCurve_HAT-P-54 b_2019-01-30.csv (4cbceb29ce3f…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)